

Radhika Achikanath Chirakkara

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Education

PhD	Thesis title : “ Magnetic field amplification in collisionless plasma ” PhD at Research School of Astronomy and Astrophysics (RSAA), Australian National University (ANU), Canberra, Australia Supervisors: Dr. Amit Seta, Prof. Christoph Federrath, and Prof. Matthew W. Kunz 03/2021-02/2025 (expected)
Bachelors & Masters	BS-MS Dual Degree with major in physics, Masters thesis title: “The small-scale dynamo in the low-Mach number regime” Indian Institute of Science Education and Research Pune, India CGPA - 9.6/10, Graduated with Distinction 08/2015-06/2020
School Education	12 th Grade, Central Board of Secondary Education (CBSE), India 94.6 % (2015); 10 th Grade, CBSE, India 10 CGPA (2013)

Research Interests

Astrophysical turbulence and magnetic fields, Dynamo theory, Plasma physics, Galaxy clusters, Kinetic and MHD simulations, Numerical methods, High-performance computing

List of Publications

Refereed publications

1. **Radhika Achikanath Chirakkara**, Christoph Federrath and Amit Seta, “A comparison of the turbulent dynamo in collisionless and collisional plasmas: from subsonic to supersonic turbulence”, (submitted, 2024)
2. **Radhika Achikanath Chirakkara**, Christoph Federrath and Amit Seta, “AHKASH: a new hybrid particle-in-cell code for simulations of astrophysical collisionless plasma”, *Monthly Notices of the Royal Astronomical Society* (2024)
3. **Radhika Achikanath Chirakkara**, Amit Seta, Christoph Federrath and Matthew W Kunz, “Critical magnetic Reynolds number of the turbulent dynamo in collisionless plasmas”, *Monthly Notices of the Royal Astronomical Society* (2024)
4. **Radhika Achikanath Chirakkara**, Christoph Federrath, Pranjal Trivedi, Robi Banerjee, “Efficient Highly Subsonic Turbulent Dynamo and Growth of Primordial Magnetic Fields”, *Physical Review Letters* (2021)

Conference proceedings

1. **Radhika Achikanath Chirakkara**, Christoph Federrath and Amit Seta, “A new ion cooling method for hybrid particle-in-cell codes”, Paper accepted for publication, (2024)

Grants, Awards and Fellowships

1. Australian Government Research Training Program (AGRTP) Stipend Scholarship (36,652 Australian Dollars (AUD) per annum, 2021-present)
2. ANU RSAA Supplementary Research Scholarship (5,000 AUD per annum, 2021-2024)
3. Member of the Astronomical Society of Australia (2023-present)
4. Zonta Club of Canberra Breakfast RSAA Scholarship, awarded in recognition of advocacy for women in STEM (3,000 AUD, 2024)

5. ANU Vice-Chancellor’s Higher Degree Research Travel Grant, ANU (3,000 AUD, 2024)
6. RSAA ANU International Travel Award (5,000 AUD, 2022-2024)
7. Australian HPC-AI Talent Program Scholarship (Independent computational grant of 100 KSU on NCI Gadi Supercomputer and 10,000 AUD travel grant, 2023)
8. ANU Full Residential Scholarship (17,027 AUD, 2023)
9. Visiting Research Scholar at the University of Hamburg, Germany (2019-2020)
10. Future Research Talent fellowship, ANU (6,000 AUD, 2019)
11. Working Internships in Science and Engineering scholarship awarded by the German Academic Exchange Service - DAAD, Germany (2018)
12. National Initiative on Undergraduate Science awarded by the Homi Bhabha Centre for Science Education, Tata Institute of Fundamental Research, India (2016-2017)
13. Kishore Vaigyanik Protsahan Yojana fellowship awarded by the Indian Institute of Science, India (2015 - 2020)

Seminars and Talks

1. Plasma Astro meeting talk at the Centre for Computational Astrophysics, Flatiron Institute, U. S. A. (Sept. 2024)
2. Talk at the University of Chicago, U. S. A. (Sept. 2024)
3. Astro Journal Club seminar, University of Rochester, U. S. A. (Aug. 2024)
4. Helsinki Astrophysics Seminar, University of Helsinki, Finland (July 2024)
5. Cosmology seminar, Leibniz Institute for Astrophysics Potsdam (AIP), Germany (July 2024)
6. Talk at the Geneva Observatory, Switzerland (July 2024)
7. Invited speaker at the Plasma Astro meeting at the Canadian Institute for Theoretical Astrophysics, Canada (Mar. 2024)
8. International Centre for Radio Astronomy Research and University of Western Australia seminar, Australia (Nov. 2023)
9. Astronomical Society of Australia – Early Career Researcher seminar, University of New South Wales, Australia (Nov. 2023)
10. Talk at the University of Sydney, Australia (Nov. 2023)
11. Talk at the Inter-University Centre for Astronomy and Astrophysics, India (Jan. 2023)
12. Talk at the Indian Institute of Technology Indore, India (Jan. 2023)
13. Talk at the Indian Institute of Science, India (Jan. 2023)
14. Seminar at the International Centre for Theoretical Sciences, India (Jan. 2023)
15. Guest lecture, Astrophysical Gas Dynamics course, RSAA (Oct. 2022)
16. Virtual Nordic Dynamo Seminar, Nordic Institute for Theoretical Physics, Sweden (Sept. 2021)

Workshops and Conferences

1. Australian National Institute for Theoretical Astrophysics workshop, Australia (Jan. 2025)
2. 24th Australasian Fluid Mechanics Conference (*contributed talk*), Australia (Dec. 2024)
3. Galaxy Clusters & Radio Relics II Meeting (*contributed talk*), Centre for Astrophysics, Harvard & Smithsonian, U. S. A. (Sept. 2024)
4. 15th International Symposium for Space Simulations (*contributed talk*) and the 16th International Workshop on the Interrelationship between Plasma Experiments in the Laboratory and in Space, Germany (Aug. 2024)
5. European Astronomical Society Annual Meeting 2024 (*contributed talks*), Italy (July 2024)
6. Australian Leadership Computing Symposium 2023 (*contributed talk*), Australia (June 2023)
7. NCI NVIDIA High-performance computing-AI Hackathon, Australia (June 2023)
8. Summer school on “Plasmas in extreme environments: from astrophysics to the laboratory” at Les Houches School of Physics (*contributed talk*), France (May 2023)
9. IAU meeting “Challenges and Innovations in Computational Astrophysics IV” (*contributed talk*) (Nov. 2022)

10. 14th International School for Space Simulations (Sept. 2022)
11. Conference on Magnetic field evolution in low density or strongly stratified plasmas (*contributed talk*), Nordic Institute for Theoretical Physics, Sweden (May - June 2022)
12. Australian National Institute for Theoretical Astrophysics workshop (*contributed talk*), Australia (Feb. 2022)
13. Kavli Institute for Astronomy and Astrophysics Forum on Gas in Galaxies for Early Career Scientists (Nov. 2021)
14. IAU Symposium 362 “Predictive Power of Computational Astrophysics as a Discovery Tool” (Nov. 2021)
15. Royal Astronomical Society Specialist Discussion Meeting on Galactic Magnetic Fields (*contributed talk*) (Oct. 2021)
16. Astronomical Society of Australia, Annual Science Meeting (*contributed talk*) and Harley Wood School of Astronomy, Australia (July 2021)

Computational Proficiency

- Lead developer of the AHKASH (“Astrophysical Hybrid Kinetic simulations with f~~A~~SH”) code
- Programming Experience: Fortran, Python, Interactive Data Language (IDL), MATLAB, Shell scripting, MPI
- Supercomputers used: Gadi, Raijin (National computing infrastructure, Australia), Setonix (Pawsey supercomputing centre, Australia), SuperMUC-NG (Leibniz Rechenzentrum, Germany) HUMMEL (University of Hamburg, Germany)
- Codes/Softwares: FLASH (Numerical code for compressible hydrodynamic and magnetohydrodynamic flow simulations), PENCIL (High-order finite-difference code for compressible fluid dynamics with magnetohydrodynamics)
- Acquainted with HEASoft and XSPEC (X-Ray Data Analysis) and have worked with astronomical data from the Solar and Heliospheric Observatory (SOHO)’s - LASCO and EIT, Astrosat and Observatory of Paris - Meudon

Teaching, Mentorship and Outreach

- Outreach event for the Zonta Club of Canberra Breakfast at the RSAA (Nov. 2024)
- Tutor and grader for PHYS1101 (Semester 2 2023, Semester 1 2024), an Undergraduate course offered by the Research School of Physics, Australian National University
- Mentor for the MSATT Astronomy program at the Mount Stromlo Observatory (2023)
- Mentoring Bachelor of Science (Honours) research students at the RSAA through the RSAA Student Buddy System (2023-2024)

Synergistic Activities/Positions

- Local organizing committee volunteer for the 24th Australasian Fluid Mechanics Conference (AFMC 2024)
- Member of Cosmology and Astronomy Researchers of Indian Nationality And Sisters (CARINAs) (since 2024)
- Organizer of 2023 Student Writing Retreat at RSAA, ANU (see RSAA newsletter on the event)
- Student ambassador for the ANU Future Research Talent Programme (2023 & 2024)
- PhD student welcome committee at RSAA (2023) and work health safety volunteer at RSAA (2022-2023)
- Co-organiser of the 2022 Mount Stromlo Student Seminar, RSAA, ANU
- Member of the Women in Astrophysical Fluids and Plasmas Forum (since 2021)
- Academic Team Member at the 10th International Olympiad on Astronomy & Astrophysics (Dec. 2016)

References

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